

Figure 1a. RL training-regime design matrix - feedback specification

Each row is a named regime. Dark cells are defining or canonical choices; light cells are common alternatives. Blank cells mean that the regime name leaves that choice open. Regime names are partial specifications, not mutually exclusive classes.

Axes A-D: feedback source, reward semantics, signal representation, and temporal granularity

Named regime	A. Feedback provenance					B. Reward semantics					C. Signal representation					D. Temporal granularity						
	Human	AI model	Current policy self	Verifier	Environment	Hybrid source	Subjective	Normative	Verifiable	Learned proxy	Hybrid semantics	Pairwise	Listwise	Unary	Scalar	Demon- stration	Constraint vector	Token	Step process	Response	Turn	Trajectory
Preference-derived alignment																						
Classical RLHF	✓					○	✓			✓	○	✓			✓					✓	○	
Online RLHF	✓					○	✓			○		✓			○				○	✓	○	
Offline RLHF	✓						✓					✓			○					✓		○
Multi-objective / constrained RLHF	✓	○		○		✓	○	○	○		✓				○		✓			✓		○
RLAIF		✓					✓	○		○		✓			○				○	✓		
Constitutional RLAIF	○	✓					○	✓		○		✓			○				○	✓		
Verifier-derived and objective-feedback regimes																						
RLVR (verifiable outcome)				✓					✓						✓		○		○	✓		○
RLVP (verifiable path penalties + outcome)[52]				✓	✓	✓			✓		○				✓		✓		✓	○	○	✓
RLVRR (verifiable reference/reward chain)[53]		○		✓		✓			✓	○	✓		○		✓		○		✓	✓		
RL from execution feedback				✓	✓	○			✓						✓		○		○		○	✓
Credit-granularity regimes																						
Outcome-reward RL	○	○		○	○	✓	○		○		✓				✓					✓		○
Process-reward RL	○	○		○		✓	○		○	○	✓				✓			○	✓	○		
Self-improving and game regimes																						
Self-rewarding RL		○	✓			○	✓			✓	○	✓			○				○	✓		
Self-play RL			✓		○	✓	○				✓	✓			○					○		✓
Environment and agentic regimes																						
Agentic RL	○	○		○	✓	✓	○		○		✓				✓		○		○		✓	✓
Agentic RLVR				✓	✓	○			✓						✓		○		○		○	✓

Organization. Rows are grouped by the part of the design vector named most directly: preference-derived alignment, verifier-derived feedback, credit granularity, self-improvement/game structure, and environment interaction. These groups overlap conceptually. For example, agentic RLVR is both verifier-derived and agentic; it is placed in the agentic cluster because its distinguishing feature is long-horizon environment interaction.

Figure 1b. RL training-regime design matrix - operational specification

The same named regimes and row order as Figure 1a, now projected onto operational choices. Dark cells are defining or canonical; light cells are common alternatives; blank cells remain unspecified.

Axes E-I: data-policy coupling, interaction structure, reward/judge status, objective composition, and exploration loop

Named regime	E. Data-policy coupling				F. Interaction structure				G. Reward / judge status				H. Objective composition					I. Exploration / data loop					
	Offline fixed	Replay off-policy	Iterative refresh	On-policy online	Single-turn	Conver-sation	Tool environment	Multi-agent self-play	Direct rule	Fixed RM	Updated RM judge	LLM-as-judge	Self-judge	Single scalar	Weighted multi-objective	Constrained	Lexico-graphic	Minimax game	Static data	Best-of- <i>N</i> rejection	Self-generation	Curriculum	Adversarial self-play
Preference-derived alignment																							
Classical RLHF	○			✓	✓	○				✓				✓	○				○	○			
Online RLHF			○	✓	○	✓				○	✓			✓	○								
Offline RLHF	✓	○			○	○				○				✓					✓				
Multi-objective / constrained RLHF	○		○	○	○	○				○	○	○			✓	✓	○		○	○			
RLAIF	○		○	○	✓	○				○	○	✓		✓					○		○		
Constitutional RLAIF	○		✓	○	○	○				○	○	✓			○	○	○				✓		
Verifier-derived and objective-feedback regimes																							
RLVR (verifiable outcome)			○	✓	○		○		✓					✓						○		○	
RLVP (verifiable path penalties + outcome)[52]		○		✓			✓		✓					○		✓	○					○	
RLVRR (verifiable reference/reward chain)[53]	○		○	✓					✓			○			✓	○			○	○			
RL from execution feedback		○		✓			✓		✓					✓	○	○				○		○	
Credit-granularity regimes																							
Outcome-reward RL	○	○		○	○	○	○		○	○	○			✓									
Process-reward RL	○		○	○	○	○	○		○	○	○	○		✓	○								
Self-improving and game regimes																							
Self-rewarding RL			✓	○	○	○					○	○	✓	○							✓		
Self-play RL			✓	○				✓			○		○	○				✓		○	○		✓
Environment and agentic regimes																							
Agentic RL		○		✓		○	✓		○	○	○			○	✓	○						○	
Agentic RLVR		○		✓			✓		✓					✓	○	○				○		○	

○

Preference/RLHF AI-feedback Verifier/objective Credit granularity Self-learning/game Agentic/environment

RLVP naming note. Here “RLVP” refers to Li and Shi’s *Penalize the Path, Reward the Outcome* regime: verifiable penalties constrain undesirable path events while an outcome reward preserves task completion. The acronym was also used in 2026 for “Reinforcement Learning with Verifiable Physics”; that domain-specific usage is not the row shown here.[52]

Figure 2a. Core online policy-gradient and value-based algorithm families

The tree separates learned-value, critic-free, group-relative, and logged-data branches. Algorithm names marked [P] come from named papers.

Online scalar-return update academic algorithm Offline/value or logged-data method solid arrow: main lineage dashed arrow: sibling estimator, granularity, or trust-region redesign shared orthogonal trunk: sibling branches from one parent [P] named

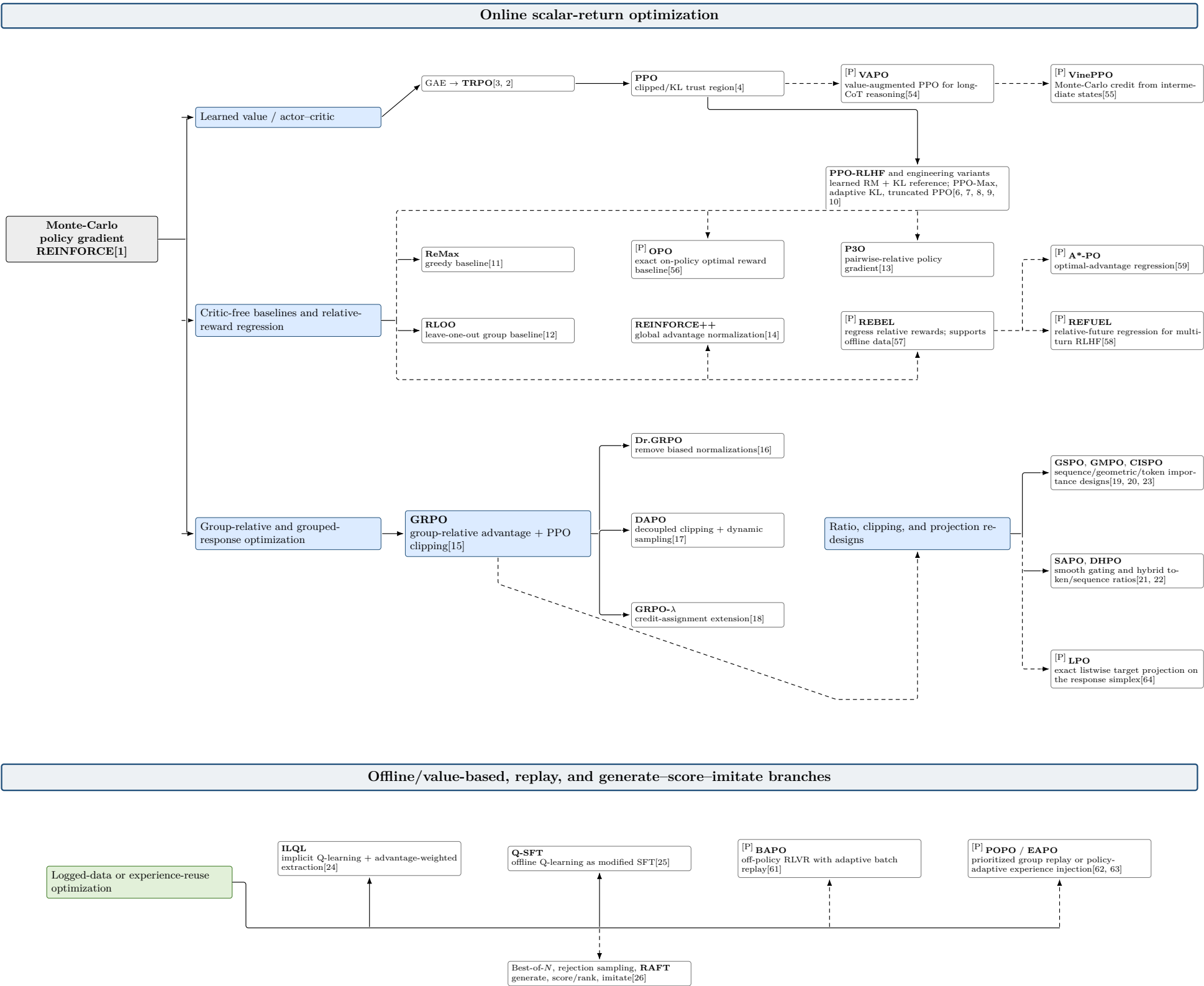


Figure 2b. Direct, iterative, and game-theoretic preference optimization

These methods consume comparisons, rankings, unary labels, or preference oracles. Standard fixed-data DPO is derived from regularized RL but does not perform an on-policy RL rollout loop.

Fixed-data/direct preference objective Iterative, self-play, or preference-game method solid arrow: main objective lineage dashed arrow: feedback-granularity or game-structure change shared orthogonal trunk: siblings with a common parent

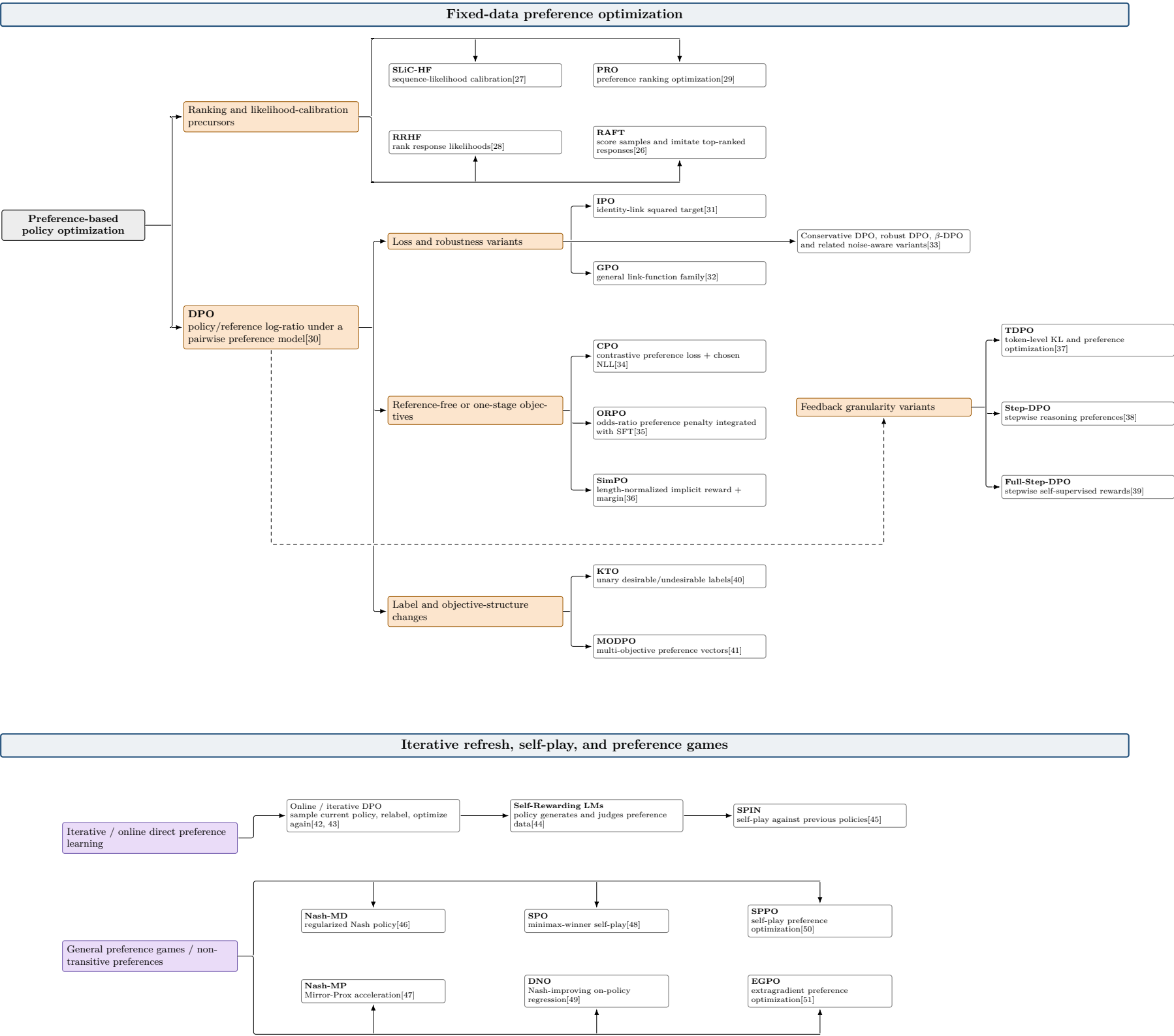


Figure 2c. Process-credit, agentic, and model-report algorithm modules

These methods modify credit assignment, trajectory collection, experience reuse, agent orchestration, or teacher consolidation. They are not all interchangeable base policy-gradient estimators.

[P] named policy-optimization paper [R] reward/credit/data module attachable to several optimizers [M] model-report-specific recipe **solid arrow**: principal lineage **dashed arrow**: module attachment or recipe dependency **dotted border**: emerging 2026 method whose influence is not yet established

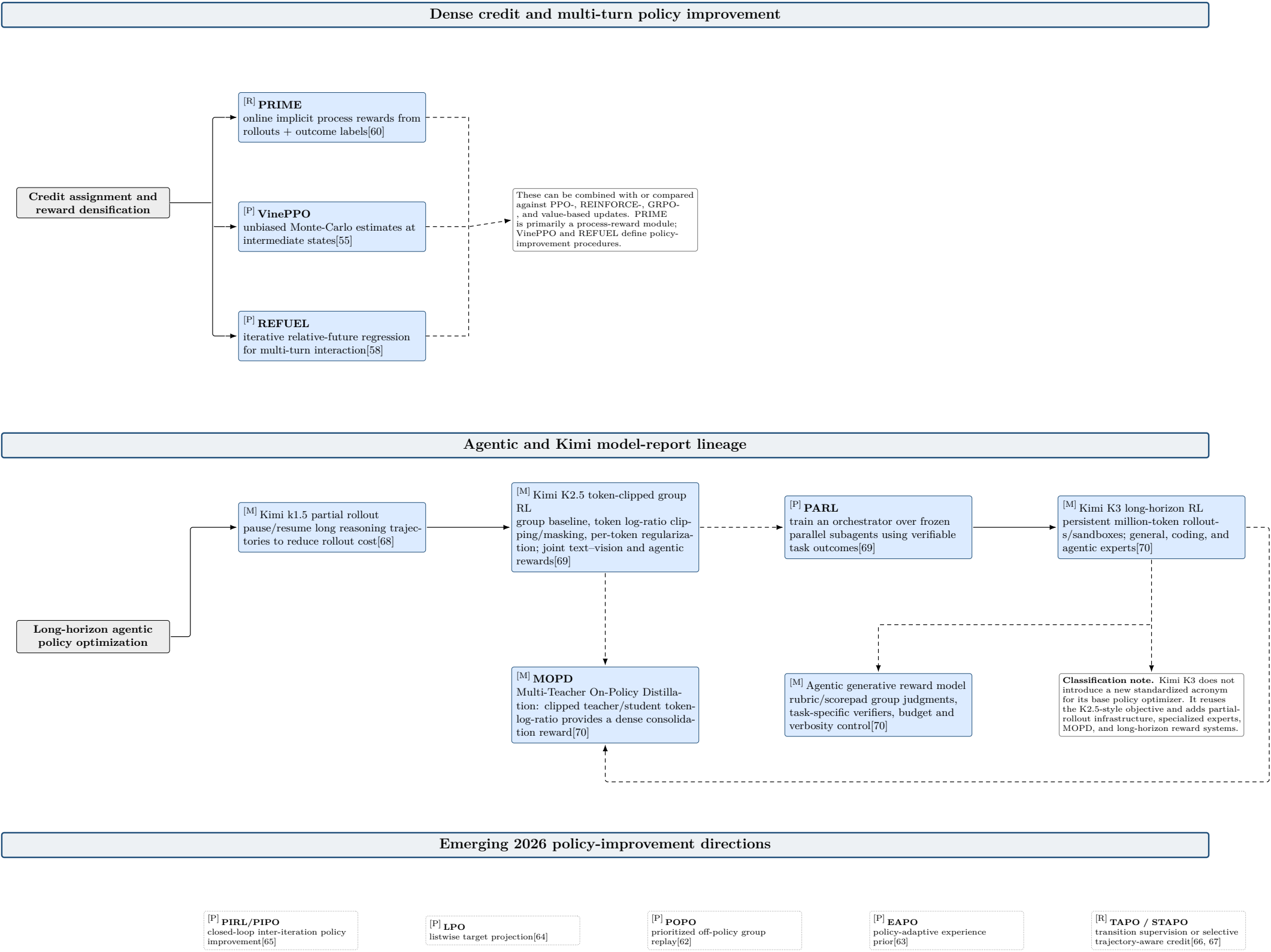


Figure 3a. Online, value, process, and agentic algorithms by signal interface

A many-to-many compatibility map. The entries describe the interface needed by the algorithm, not the historical regime in which it was first demonstrated.

A

△

Algorithm family	Scalar outcome	Scalar process	Pair/list preference	Trajectory/environment	Verifier	Learned RM/judge	Fixed offline data	Fresh on-policy	Stale/replay data	Grouped samples	Long horizon	Multi-agent
PPO / actor-critic / VAPO	D	D	A	D	D	D	△	D	△	–	D	A
VinePPO	D	D	A	D	D	A	△	D	–	A	D	–
ReMax / RLOO / REINFORCE++ / OPO	D	D	A	D	D	D	△	D	△	A	A	–
P3O / REBEL / A*-PO	A	△	D	A	A	A	A	D	A	D	△	–
REFUEL	D	A	A	D	A	D	A	D	A	–	D	–
GRPO / Dr.GRPO / DAPO / GRPO-λ	D	D	A	A	D	D	△	D	△	D	A	–
GSPO / GMPO / CISPO / SAPO / DHPO	D	D	A	D	D	D	△	D	A	A	D	–
LPO	D	A	A	A	D	A	△	D	–	D	△	–
ILQL / Q-SFT	D	D	A	D	D	A	D	–	D	–	D	–
BAPO / POPO / EAPO	D	A	A	A	D	A	D	A	D	D	A	–
[R] PRIME	A	D	–	A	D	A	△	D	△	A	D	–
[M] Kimi K2.5/K3 token-clipped group RL	D	D	A	D	D	D	–	D	D	D	D	A
PARL	D	A	A	D	D	A	–	D	△	A	D	D
[M] MOPD	A	D	–	A	–	D	–	D	A	A	D	A
PIRL/PIPO and TAPO/STAPO	D	D	A	D	D	A	A	D	A	A	D	A

Reading examples. A verifier is native for GRPO but can also train PPO, RLOO, OPO, Kimi-style group RL, or an agent orchestrator. Pairwise-only feedback is native for P3O/REBEL but needs an RM or pair-to-scalar adapter for ordinary PPO/GRPO. Kimi K3 is represented as a model-report recipe, not as a newly standardized optimizer.

Figure 3b. Direct, iterative, and game-theoretic preference algorithms by signal interface

These rows primarily consume comparisons, rankings, unary labels, or preference oracles; verifier and scalar rewards usually require candidate-to-pair conversion.

A△

Algorithm family	Scalar outcome	Scalar process	Pairwise	Listwise/ranking	Unary label	Verifier	Non-transitive oracle	Fixed offline data	Fresh policy data	Grouped samples	Long horizon
Best-of- N / rejection sampling / RAFT	D	A	A	D	A	D	A	D	A	D	–
SLiC-HF / RRHF / PRO	A	A	D	D	A	A	A	D	△	–	–
DPO / IPO / GPO	A	A	D	A	–	A	D	D	△	–	–
CPO / ORPO / SimPO	A	A	D	A	–	A	A	D	△	–	–
KTO	A	A	A	A	D	A	A	D	△	–	–
MODPO	A	A	D	A	A	A	A	D	△	–	–
TDPO / Step-DPO / Full-Step-DPO	A	D	D	A	–	A	A	D	△	–	A
Online/iterative DPO; Self-Rewarding; SPIN	A	A	D	D	A	D	D	A	D	D	A
Nash-MD / Nash-MP / SPO / DNO / SPPO / EGPO	A	–	D	A	–	A	D	A	D	D	△

Core conclusion. Regime and optimizer are separate dimensions, but they combine through interfaces rather than arbitrarily: feedback representation, data freshness, grouped sampling, off-policy correction, interaction horizon, and judge stationarity constrain which combinations are direct and which require conversion.

Figure 4a. Named regimes: preference, verifier, and credit compositions

Fixed two-column layout; card colors match the Figure 1 clusters. Interface lists are non-exclusive.

Preference/RLHF AI feedback Verifier/objective Credit granularity

Classical / online / offline RLHF

Signature: human feedback; usually subjective pairwise labels; optionally distilled into a scalar RM; response, turn, or trajectory credit; offline, iterative, or on-policy variants.

Interfaces: PPO/VAPO, ReMax/RLOO/OPO, REINFORCE++, GRPO-style updates after scalarization; DPO-family losses consume comparisons directly.

RLAIF / constitutional RLAIF

Signature: AI-generated critiques, principles, preferences, or judge scores; subjective or normative semantics; fixed, iterative, or online labels.

Interfaces: reward-model PPO/RLOO/GRPO, direct DPO/IPO/ORPO/SimPO, online DPO, and self-rewarding loops.

RLVP and RLVR

RLVP: verifiable path penalties plus task-outcome reward for costly or irreversible trajectories.[52]

RLVR: reference-derived reward chains combining deterministic checks and model-based verification for open-ended generation.[53]

Interfaces: trajectory-aware scalar-return optimizers; pair/rank methods after conversion.

Multi-objective / constrained RLHF

Signature: multiple human, AI, verifier, safety, cost, or style dimensions; scalarized, constrained, lexicographic, or Pareto objectives.

Interfaces: constrained policy gradient, Lagrangian PPO/REINFORCE/GRPO, MODPO, or staged optimization.

RLVR: verifiable outcome rewards

Signature: deterministic or execution-based correctness for completed outputs; usually fresh on-policy samples.

Interfaces: PPO, RLOO/OPO, GRPO/DAPO, GSPO-family methods, LPO, Kimi-style group RL, BAPO/POPO/EAPO; DPO requires score-to-pair conversion.

Outcome-reward and process-reward RL

Signature: terminal response/episode rewards versus token-, step-, turn-, or transition-level credit.

Interfaces: PPO/VAPO/VinePPO, PRIME, GRPO-family methods with dense shaping, or TDPO/Step-DPO/Full-Step-DPO for fixed stepwise preferences.

Legend. “Signature” specifies the training regime; “Interfaces” lists algorithm families that consume the signal directly or after a stated conversion. No card defines an exclusive optimizer.

Figure 4b. Named regimes: self-improving and agentic compositions

Fixed two-column layout; card colors match the Figure 1 clusters.

Self-learning/game

Agentic/environment

^[M] model-report-specific recipe

Self-learning / self-rewarding RL

Signature: the current policy generates tasks, answers, critiques, labels, judges, or curricula; the feedback distribution changes with the learner.
Interfaces: iterative DPO, Self-Rewarding LMs, SPIN, policy gradient with a self-trained judge, or verifier RL on self-generated tasks.

Agentic RL and agentic RLVR

Signature: multi-turn tool/environment interaction; trajectory returns, execution checks, costs, and long-horizon credit; on-policy or replay data.
Interfaces: actor–critic, REFUEL, RLOO/GRPO-style trajectory updates, offline value RL, PARL, and transition-aware credit modules.

Self-play / preference-game RL

Signature: current and previous policies or opponents produce relative wins/preferences; a globally transitive scalar reward may not exist.
Interfaces: P3O, REBEL, Nash-MD/Nash-MP, SPO, DNO, SPPO, and EGPO.

^[M] Kimi K2.5/K3 agentic post-training

Signature: grouped and partially stale long-horizon rollouts; rule/verifier, generative-judge, visual, budget, and environment rewards; persistent tool sandboxes.
Interfaces: K2.5 token-clipped group objective; PARL for trainable orchestration over frozen workers; K3 partial rollout, specialized RL experts, MOPD, and million-token agentic RL.^[69, 70]

Core conclusion: regime $\perp$ optimizer conceptually, but composition is mediated by the signal interface, data freshness, off-policy correction, group structure, judge stationarity, temporal credit horizon, and whether the trainable object is one policy or an orchestrator over multiple agents.

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